SHRINCS specification

Blockstream Research

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Abstract

This document specifies **SHRINCS** (Stateful Hash-based Robust Integrated NIST-Compatible Signatures), a post-quantum signature scheme optimized for Bitcoin. SHRINCS combines efficient stateful hash-based signatures (unbalanced XMSS with WOTS+C) for normal operation with a stateless fallback (SPHINCS+C variant) for scenarios when state management failed. The scheme achieves minimal 324-byte signatures during normal stateful operation while maintaining the robustness of stateless signatures as a backup when state is corrupted or lost.

1 Motivation

Current Bitcoin signature schemes (ECDSA and Schnorr over secp256k1) are vulnerable to quantum attacks via Shor's algorithm. Hash-based signature schemes offer post-quantum security relying only on the security of cryptographic hash functions, which are believed to resist quantum attacks apart from Grover's quadratic speedup.

The key challenges for deploying hash-based signatures in Bitcoin are:

1. Signature size: NIST-standardized SLH-DSA signatures range from 7KB to 50KB, significantly larger than current 64-byte Schnorr signatures.
2. Stateful vs. Stateless tradeoff: Stateful schemes (XMSS) offer small signatures but require careful state management. Stateless schemes (SPHINCS+) are robust but produce large signatures.
3. Backup compatibility: Bitcoin users expect to restore wallets from seed phrases. Pure stateful schemes break this model since state cannot be statically backed up.

SHRINCS addresses these challenges by:

- Using efficient stateful signatures ($\approx$ 324 bytes for $q = 1$) during normal operation.
- Falling back to stateless signatures ($\approx$ 3-4 KB) when state is lost or corrupted.
- Allowing static seed backup while maintaining security guarantees.

2 Notation

Throughout this specification:

`||` denotes byte string concatenation

`[a:b]` denotes a substring from byte index `a` to `(b-1)`

`[0:a]` denotes first `a` bytes (truncation)

`←$` uniform random sampling

`n` is the security parameter in bytes

`H` denotes a cryptographic hash function

`Tw` denotes a tweakable hash function

`PRF` denotes a pseudorandom function

$\perp$ used for failure returns

3 Overview

SHRINCS is designed in the way it can operate in two modes: **stateful** and **stateless**. Stateful mode is represented by the instance of Unbalanced XMSS (UXMSS) with a root public key `PK.sf`. An optimized variant of SLH-DSA is used to support the stateless operating mode (which is represented by the public key `PK.sl`). The root public key is a hashed concatenation of (`PK.sf`, `PK.sl`), which is being converted to the Bitcoin address.

During normal operation with intact state, the user generates signatures according to efficient stateful path. Each leaf of the UXMSS is represented by WOTS+C one time signature instance. UXMSS is right-skewed, which makes the first stateful signature (`q=1`) the cheapest from the verification and size perspective (324 bytes) and adds the constant size (16 bytes) and one additional hashing operation for each next stateful signature. A right-skewed UXMSS is a UXMSS construction in which every internal node has a one-time signature leaf as its left child and a UXMSS subtree as its right child, yielding a recursively right-branching authentication tree (as shown on the Figure 1).

If the state is corrupted (or the limit for stateful operations is met), the signature can continue operating with stateless mode. Stateless mode utilizes an optimized SLH-DSA variant with WOTS+ signatures in regular XMSS trees on each layer in the hypertree and FORS+C for few-time signature scheme. The general construction of SHRINCS signature is presented on the Figure 1.

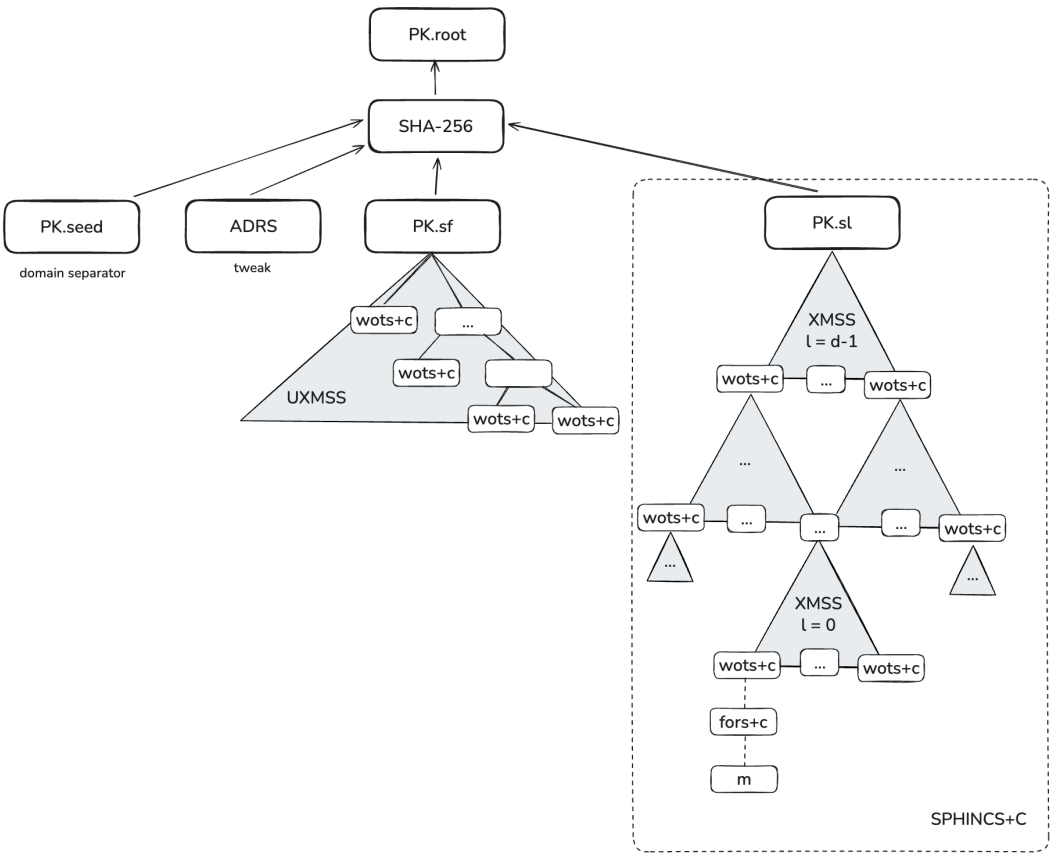

Figure 1: SHRINCS architecture

4 Parameters and Primitives

We define SHRINCS-SHA2-128+C as a SHRINCS instance preferable for Bitcoin. It operates with the following set of parameters:

Parameter	Value	Description
`H`	`SHA-256`	The underlying hash function
`OTS`	`WOTS+C`	The underlying one-time signature
`FTS`	`FORS+C`	The underlying few-time signature
`n`	16	Security parameter (bytes)
`w`	256	Winternitz parameter (8 bits per chain)
`len`	16	Message digest length, the number of WOTS+C chains
`Swn`	2040	Target sum for WOTS+C
`hsf`	15	Maximum stateful tree height
`hsl`	40	Maximum stateless hypertree height
`d`	5	Stateless hypertree layers
`a`	22	FORS+C tree height
`k`	6	Number of FORS+C trees
`c`	4	Counter size for WOTS+C grinding (bytes)
`R`	32	Randomness (bytes)

4.1 SHRINCS Key Material

SHRINCS keys are derived from a single random seed:

```
seed ←$ {0,1}^(3n)
```

Seed allows to derive a private key `SK ← (SK.seed, SK.prf, PK.seed, PK.sf, PK.sl, PK.root)`:

```
SK.seed ← seed[0:n]  // Seed for key derivation
SK.prf ← seed[n:2n]  // Secret needed for PRF_msg
PK.seed ← seed[2n:3n] // Domain separator for different SHRINCS key pairs
PK.sf ← compute_stateful_root(PK.seed, ADRS, SK.seed) // Public stateful key, see Section X
PK.sl ← compute_stateless_root(PK.seed, ADRS, SK.seed) // Public stateless key, see Section X
PK.root ← H(PK.seed, ADRS, PK_sf, PK_sl) // Root public key
```

where `ADRS` is a context address (see Section 5.1).

The size of the secret key is `6n` bytes. A public key consists of two values `PK ← (PK.seed, PK.root)` and has a size of `2n` bytes.

4.2 Hash Function

SHRINCS uses SHA-256 truncated to `n` bits for all internal hash operations.

```
H(m) = SHA-256(m)[0:n]
```

4.3 Tweakable Hash Function

To prevent multi-target attacks and ensure domain separation between the various components, we employ the tweakable hash function construction defined in FIPS 205 [**fips-205**]:

```
Tw(PK.seed, ADRS, m) = H(PK.seed || ADRS || m)
```

This usage of tweaked hashing is critical. It ensures that a hash computed at one position in the Merkle tree cannot be substituted for a hash at another position, effectively mitigating generic tree-grafting attacks.

4.4 Pseudorandom Functions

We use PRF for deterministic key derivation:

```
PRF(PK.seed, ADRS, SK.seed) = Tw(PK.seed, ADRS, SK.seed)
```

Additionally we use a pseudorandom function to generate the randomizer `R` for the randomized hashing of the message to be signed.

```
PRF_msg(SK.prf, PK.seed, opt_rand, m) = HMAC-SHA-256(SK.prf, PK.seed || opt_rand || m),
```

where `opt_rand` is an optional randomness (`n` bytes), can be `PK.seed` for deterministic signing.

4.5 Message Hash Function

To generate the digest of the message to be signed we use the following function:

```
H_msg(ADRS, R, PK.seed, PK.root, m) = H(ADRS || R || PK.seed || PK.root || m)
```

5 SHRINCS-SHA2-128s Specification

5.1 Tweak Structure and Domain Separation

Tweaks are structured as follows to ensure uniqueness across all scheme components:

```
ADRS = layer || tree_address || type || words
```

where:

`layer`: (4 bytes) layer in hypertree (`0` = bottom, `d-1` = max)

`tree_address`: (12 bytes) tree address within layer

`type`: (4 bytes) address type

`words`: (12 bytes) type-dependent 12 bytes

A total `ADRS` size is 32 bytes.

5.1.1 Address Types

`0x00`: Stateful WOTS+C chains (`SF_WOTS_CHAIN`)

`0x01`: Stateful tree nodes (`SF_TREE`)

`0x02`: Stateless WOTS+C chains (`SL_WOTS_CHAIN`)

`0x03`: Stateless tree nodes (`SL_TREE`)

`0x04`: FORS (`SL_FORS`)

`0x05`: Message hash (`H_MSG`)

`0x06`: Pseudorandom function (`PRF`)

`0x07`: Root public key (`ROOT`)

Summarizing, `(0x00, 0x01)` are stateful only types, `(0x02, 0x03, 0x04)` are stateless only and `(0x05, 0x06, 0x07)` are shared types.

5.1.2 Address Manipulation Functions

```
setLayerAddress(ADRS, layer):
    ADRS[0:4] ← toByte(layer, 4)

setTreeAddress(ADRS, tree_addr):
    ADRS[4:16] ← toByte(tree_addr, 12)

setTypeAndClear(ADRS, type):
    ADRS[16:20] ← toByte(type, 4)
    ADRS[20:32] ← toByte(0, 12)

setKeyPairAddress(ADRS, keypair):
    ADRS[20:24] ← toByte(keypair, 4)

setChainAddress(ADRS, chain):
    ADRS[24:28] ← toByte(chain, 4)

setHashAddress(ADRS, hash):
    ADRS[28:32] ← toByte(hash, 4)

setTreeHeight(ADRS, height):
    ADRS[24:28] ← toByte(height, 4)

setTreeIndex(ADRS, index):
    ADRS[28:32] ← toByte(index, 4)
```

Note that `setTreeHeight` and `setChainAddress` both write to bytes `[24:28]`. These operations are context-dependent so there mustn't be a conflict.

Oleks: TODO: type-dependent words

5.2 WOTS+C

The standard Winternitz One-Time Signature (WOTS+) is the building block of most modern hash-based schemes. It works by encoding the message into a series of small integers and using them to iterate hash chains. A significant portion of a WOTS+ signature size (approx 20%) is dedicated to "Checksum," which is necessary to prevent a forgery attack where an adversary extends the hash chains forward.

WOTS+C (Compressed WOTS+) eliminates the transmission of this checksum entirely, achieving a smaller signature size. In WOTS+, the checksum C is calculated from the message digest D.

$$C = \sum_{i=1}^{l} (w - 1 - d_i)$$

This checksum is then signed using additional chains. In WOTS+C, instead of calculating a checksum for a fixed message digest, the signer modifies the message digest until it possesses a specific, predetermined checksum property. This is done by appending a counter `count` and a random salt `R` to the message before hashing. The signer increments cnt until the resulting hash digest D satisfies a global target condition.

5.2.1 The Target Condition

For SHRINCS-SHA2-128s+C, we use the Winternitz parameter $w = 256$. This means each chain processes 8 bits (1 byte) of the digest. The message digest length is 128 bits (16 bytes), so $l = 16$. The values d_i are treated as integers in $0 \ldots (w - 1)$.

We define the expected value of the checksum. If the hash output is uniform, the expected value of a single digit is $(w - 1)/2 = 127.5$. For 16 digits, the expected sum is $16 \times 127.5 = 2040$.

The condition for a valid WOTS+C message digest D is:

$$\sum_{i=0}^{15} d_i = 2040$$

By enforcing this condition during signing (including grinding), the verifier can simply assume the checksum is 2040 without receiving it. If the reconstructed digest does not sum to 2040, the signature is invalid.

5.2.2 Parameters

Parameter	SHRINCS-SHA2-128s
`n`	128 bits (16 bytes)
`w`	256
`len`	16 chains
`Swn`	2040
`c`	4 bytes

5.2.3 Base-w Representation

```
Algorithm: base_w(m, w, out_len) // this is generic algorithm, for w=256 it's much more simpler
Input:
    m: byte string
    w: Winternitz parameter
    out_len: output length
Output:
    b_w: array of out_len integers in [0, w-1]

1. in_idx ← 0
2. bits ← 0
3. total ← 0
4. b_w ← []
5. for i from 0 to out_len - 1:
    a. if bits == 0:
        total ← m[in_idx]
        in_idx ← in_idx + 1
        bits ← 8
    b. bits ← bits - log2(w)
    c. b_w[i] ← (total >> bits) AND (w - 1)
6. return b_w
```

For `w=256`:

```
Algorithm: base_w(m, out_len)
Input:
    m: byte string of length out_len
    out_len: output length
Output:
    b_w: array of out_len integers in [0, 255]

1. b_w ← []
2. for i from 0 to out_len - 1:
    a. b_w[i] ← m[i]
3. return b_w
```

5.2.4 Chain Function

```
Algorithm: chain(m, start, steps, PK.seed, ADRS)
Input:
    m: n-byte input
    start: starting index in chain
    steps: number of iterations
    PK.seed: public seed
    ADRS: tweak
Output:
    n-byte chain output

1. tmp ← m
2. for i from start to start + steps - 1:
    a. setHashAddress(ADRS, i)
    b. tmp ← Tw(PK.seed, ADRS, tmp)
3. return tmp
```

5.2.5 Key Generation

```
Algorithm: wots_PKgen(SK.seed, PK.seed, ADRS, keypair)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
    keypair: keypair index for this WOTS+ instance
Output:
    pk: compressed WOTS+C public key (n bytes)

1. tmp ← []
2. for i from 0 to len - 1:
    a. setTypeAndClear(ADRS, PRF)
    b. setKeyPairAddress(ADRS, keypair)
    c. setChainAddress(ADRS, i)
    d. sk_i ← PRF(PK.seed, ADRS, SK.seed)
    e. setTypeAndClear(ADRS, SF_WOTS_CHAIN)
    f. setKeyPairAddress(ADRS, keypair)
    g. setChainAddress(ADRS, i)
    h. tmp[i] ← chain(sk_i, 0, w - 1, PK.seed, ADRS)
3. setTypeAndClear(ADRS, SF_TREE)
4. setKeyPairAddress(ADRS, keypair)
5. pk ← Tw(PK.seed, ADRS, tmp[0] || ... || tmp[len-1])
6. return pk
```

5.2.6 Messsage Digest Grinding

Oleks: To consider if we need to mention counter in Section 4.5

```
Algorithm: wots_grind(m, R, PK.seed, PK.root, ADRS)
Input:
    m: message to sign
    R: randomness
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    msg: array of len integers
```

```
    ctr: final counter value used

1. setTypeAndClear(ADRS, H_MSG)
2. for ctr from 0 to 2^32 - 1:
    a. digest ← H_msg(ADRS, R, ctr, PK.seed, PK.root, m)
    b. msg ← base_w(digest, len)
    c. sum ← 0
    d. for i from 0 to len - 1:
        sum ← sum + msg[i]
    e. if sum == Swn:
        return (msg, ctr)
3. abort with error  // R must be resampled
```

Oleks: to add expected counter value for different approaches

Additionally we need to define a func which allows to calculate the digest without grinding (by providing the counter value)

```
Algorithm: wots_digest(m, R, ctr, PK.seed, PK.root, ADRS)
Input:
    m: message
    R: randomness from signature
    ctr: counter from signature
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    msg: array of len integers
    valid: boolean indicating if sum == Swn

1. setTypeAndClear(ADRS, H_MSG)
2. digest ← H_msg(ADRS, R, ctr, PK.seed, PK.root, m)
3. msg ← base_w(digest, len)
4. sum ← 0
5. for i from 0 to len - 1:
    sum ← sum + msg[i]
6. valid ← (sum == Swn)
7. return (msg, valid)
```

5.2.7 Signature Generation

```
Algorithm: wots_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, keypair)
Input:
    m: message
    SK.seed: secret seed
    SK.prf: PRF key for randomness
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
    keypair: keypair index for this WOTS+ instance
Output:
    sig: WOTS+C signature (R || ctr || chain_values)

1. opt_rand ← random(n) or PK.seed  // for randomized or deterministic signing
2. R ← PRF_msg(SK.prf, PK.seed, opt_rand, m)
3. (msg, ctr) ← wots_grind(m, R, PK.seed, PK.root, ADRS)
4. sig ← R || toByte(ctr, c)
5. for i from 0 to len - 1:
```

```
    a. setTypeAndClear(ADRS, PRF)
    b. setKeyPairAddress(ADRS, keypair)
    c. setChainAddress(ADRS, i)
    d. sk_i ← PRF(PK.seed, ADRS, SK.seed)
    e. setTypeAndClear(ADRS, SF_WOTS_CHAIN)
    f. setKeyPairAddress(ADRS, keypair)
    g. setChainAddress(ADRS, i)
    h. sig ← sig || chain(sk_i, 0, msg[i], PK.seed, ADRS)
6. return sig
```

5.2.8 Signature Verification (Public Key Recovery)

```
Algorithm: wots_pkFromSig(sig, m, PK.seed, PK.root, ADRS, keypair)
Input:
    sig: WOTS+C signature
    m: signed message
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
    keypair: keypair index for this WOTS+ instance
Output:
    pk: recovered public key (n bytes), or $\bot$ if invalid

1. R ← sig[0:R_SIZE]
2. ctr ← sig[R_SIZE:R_SIZE + c]
3. chains_start ← R_SIZE + c
4. (msg, valid) ← wots_digest(m, R, ctr, PK.seed, PK.root, ADRS)
5. if not valid:
    return $\bot$
6. tmp ← []
7. for i from 0 to len - 1:
    a. setTypeAndClear(ADRS, SF_WOTS_CHAIN)
    b. setKeyPairAddress(ADRS, keypair)
    c. setChainAddress(ADRS, i)
    d. sig_i ← sig[chains_start + i*n : chains_start + (i+1)*n]
    e. tmp[i] ← chain(sig_i, msg[i], w - 1 - msg[i], PK.seed, ADRS)
8. setTypeAndClear(ADRS, SF_TREE)
9. setKeyPairAddress(ADRS, keypair)
10. pk ← Tw(PK.seed, ADRS, tmp[0] || ... || tmp[len-1])
11. return pk
```

5.3 UXMSS. The Stateful Tree

Standard XMSS uses a balanced binary tree, which minimizes the maximum path length but imposes a logarithmic cost on every signature. For `N` signatures, every signature has size proportional to `log_2(N)`. SHRINCS utilizes an Unbalanced Merkle Tree (specifically a "Right-Skewed" tree) to optimize for the typical Bitcoin use case: a wallet that performs very few transactions in its lifecycle (we are targeting `N<=10`).

In this topology the `q`-th signature requires a path of length `q`. This linear growth is superior to logarithmic growth for small `q`. For `q=1`, the path is just 16 bytes. Even for `q=10`, the path is $10 \times 16 = 160$ bytes, which is extremely small compared to the fixed ≈ 2.5KB of a balanced tree or the ≈ 8KB of SLH-DSA.

5.3.1 UXMSS Tree Structure

Oleks: I hope it's not confusing we have layers numeration different for stateless and statefull instance
The UXMSS tree is constructed recursively as follows:

- Level 0 is the root public key `PK.sf`.
- Level `i` (internal):
  - Left child: WOTS+C public key for signature `q=i`
  - Right child: Root of subtree for signatures `q>i`
- Level `hsf` (bottom)
  - Left child: WOTS+C public key for signature `q=hsf`
  - Right child: WOTS+C public key for signature `q=hsf+1`

5.3.2 UXMSS Authentication Path

For the `q`-th signature (`q` = count after signing), the authentication path contains `q` nodes:

```
auth_path = [node_0, node_1, ..., node_q-1]
```

Root computation from auth path:

```
Algorithm: root_from_auth(pk, auth, q, PK.seed, ADRS)
Input:
    pk: WOTS+C public key for signature q
    auth: authentication path [auth[0], ..., auth[q-1]] from bottom to top
    q: signature number
    PK.seed: public seed
    ADRS: tweak
Output:
    root: computed tree root (PK.sf)

1. node ← pk
2. setTypeAndClear(ADRS, SF_TREE)
3. if q <= hsf:
    for i from 0 to q - 1:
        a. level ← q - i
        b. setTreeHeight(ADRS, level)
        c. setTreeIndex(ADRS, 0)
        d. if i == 0:
            node ← Tw(PK.seed, ADRS, node || auth[i])
        e. else:
            node ← Tw(PK.seed, ADRS, auth[i] || node)
4. else:  // q = h_sf + 1
    for i from 0 to h_sf - 1:
        a. level ← h_sf - i
        b. setTreeHeight(ADRS, level)
        c. setTreeIndex(ADRS, 0)
        d. if i == 0:
            node ← Tw(PK.seed, ADRS, auth[i] || node)
        e. else:
            node ← Tw(PK.seed, ADRS, auth[i] || node)
5. return node
```

5.4 FORS+C

FORS (Forest of Random Subsets) is a few-time signature scheme used at the bottom of the stateless hypertree. FORS allows to sign a limited number of messages, when the security degrades with each use. Standard FORS signs a message by revealing `k` secret values, one from each of `k` binary trees of height `a`. The message digest determines which leaf to reveal from each tree. FORS+C applies the same checksum grinding optimization as WOTS+C to reduce signature size.

5.4.1 FORS Structure

FORS consists of `k` binary trees, each of height `a`:

- Each tree has `2â` leaves
- Each leaf is derived from `SK.seed` using PRF
- The message digest is split into `k` chunks of `a` bits each
- Each chunk selects one leaf index from the corresponding tree

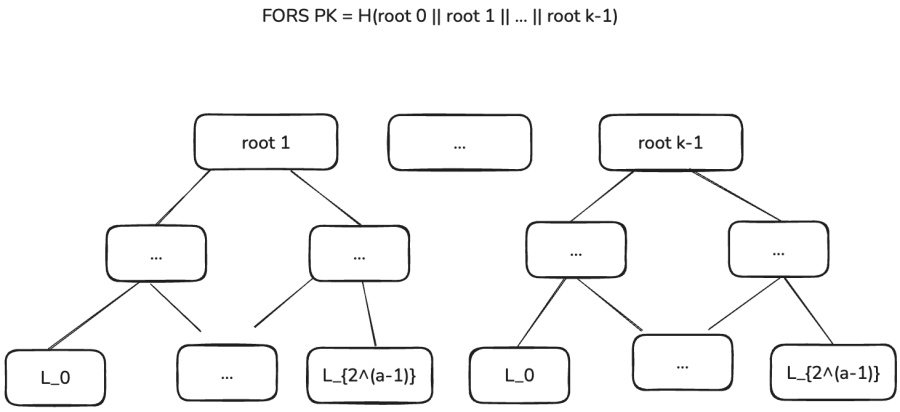

Figure 2: FROST architecture

5.4.2 Parameters

Parameter	SHRINCS-SHA2-128s
`n`	128 bits (16 bytes)
`k`	6
`a`	22

5.4.3 Grinding Condition

```
digest = H_msg(ADRS, R, ctr, PK.seed, PK.root, m)
condition: digest[last a bits] == 0  (equivalently: indices[k-1] == 0)
```

Cost is expected to be $\approx 2^a$ iterations.

5.4.4 Message to Indices

```
Algorithm: fors_msg_to_indices(digest)
Input:
    digest: message digest (ceil(k × a / 8) = 17 bytes)
Output:
    indices: array of k integers in [0, 2^a - 1]
```

```
1. indices ← []
2. for i from 0 to k - 1:
    a. indices[i] ← extract_bits(digest, i × a, a)
3. return indices
```

5.4.5 Grinding

```
Algorithm: fors_grind(m, R, PK.seed, PK.root, ADRS)
Input:
    m: message to sign
    R: randomness
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    indices: array of k integers
    ctr: counter value

1. setTypeAndClear(ADRS, H_MSG)
2. for ctr from 0 to 2^32 - 1:
    a. digest ← H_msg(ADRS, R, ctr, PK.seed, PK.root, m)
    b. indices ← fors_msg_to_indices(digest)
    c. if indices[k-1] == 0:
        return (indices, ctr)
3. abort with error
```

5.4.6 Secret Key Generation

```
Algorithm: fors_SKgen(SK.seed, PK.seed, ADRS, tree_idx, leaf_idx)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
    tree_idx: tree index (0 to k-1)
    leaf_idx: leaf index within tree
Output:
    sk: secret value (n bytes)

1. setTypeAndClear(ADRS, PRF)
2. setKeyPairAddress(ADRS, tree_idx)
3. setTreeIndex(ADRS, leaf_idx)
4. sk ← PRF(PK.seed, ADRS, SK.seed)
5. return sk
```

5.4.7 TreeHash

```
Algorithm: fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, target_height, start_idx)
Input:
    SK.seed, PK.seed: seeds
    ADRS: tweak
    tree_idx: tree index (0 to k-1)
    target_height: height of output node
    start_idx: leftmost leaf index in subtree
Output:
    node: hash value (n bytes)
```

```
1. if target_height == 0:
    a. sk ← fors_SKgen(SK.seed, PK.seed, ADRS, tree_idx, start_idx)
    b. setTypeAndClear(ADRS, SL_FORS)
    c. setKeyPairAddress(ADRS, tree_idx)
    d. setTreeHeight(ADRS, 0)
    e. setTreeIndex(ADRS, start_idx)
    f. return Tw(PK.seed, ADRS, sk)
2. left ← fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, target_height-1, start_idx)
3. right ← fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, target_height-1, start_idx +
           2^(target_height-1))
4. setTypeAndClear(ADRS, SL_FORS)
5. setKeyPairAddress(ADRS, tree_idx)
6. setTreeHeight(ADRS, target_height)
7. setTreeIndex(ADRS, start_idx >> target_height)
8. return Tw(PK.seed, ADRS, left || right)
```

5.4.8 Signature generation

```
Algorithm: fors_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS)
Input:
    m: message
    SK.seed: secret seed
    SK.prf: PRF key
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    sig: FORS+C signature

1. opt_rand ← random(n) or PK.seed
2. R ← PRF_msg(SK.prf, PK.seed, opt_rand, m)
3. (indices, ctr) ← fors_grind(m, R, PK.seed, PK.root, ADRS)
4. sig ← R || toByte(ctr, c)
5. for i from 0 to k - 2:
    a. sk_i ← fors_SKgen(SK.seed, PK.seed, ADRS, i, indices[i])
    b. auth_i ← fors_auth_path(SK.seed, PK.seed, ADRS, i, indices[i])
    c. sig ← sig || sk_i || auth_i
6. sk_last ← fors_SKgen(SK.seed, PK.seed, ADRS, k-1, 0)
7. root_last ← fors_treehash(SK.seed, PK.seed, ADRS, k-1, a, 0)
8. sig ← sig || sk_last || root_last
9. return sig
```

5.4.9 Public Key Recovery

```
Algorithm: fors_pkFromSig(sig, m, PK.seed, PK.root, ADRS)
Input:
    sig: FORS+C signature
    m: message
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    pk: FORS public key (n bytes), or $\bot$ if invalid

1. R ← sig[0 : R_SIZE]
2. ctr ← sig[R_SIZE : R_SIZE + c]
```

```
3. setTypeAndClear(ADRS, H_MSG)
4. digest ← H_msg(ADRS, R, ctr, PK.seed, PK.root, m)
5. indices ← fors_msg_to_indices(digest)
6. if indices[k-1] != 0:
    return $\bot$

7. roots ← []
8. offset ← R_SIZE + c
9. for i from 0 to k - 2:
    a. sk_i ← sig[offset : offset + n]
    b. auth_i ← sig[offset + n : offset + n + a × n]
    c. offset ← offset + n + a × n
    d. setTypeAndClear(ADRS, SL_FORS)
    e. setKeyPairAddress(ADRS, i)
    f. setTreeHeight(ADRS, 0)
    g. setTreeIndex(ADRS, indices[i])
    h. node ← Tw(PK.seed, ADRS, sk_i)
    i. idx ← indices[i]
    j. for h from 0 to a - 1:
        setTreeHeight(ADRS, h + 1)
        setTreeIndex(ADRS, idx >> 1)
        if (idx AND 1) == 0:
            node ← Tw(PK.seed, ADRS, node || auth_i[h × n : (h+1) × n])
        else:
            node ← Tw(PK.seed, ADRS, auth_i[h × n : (h+1) × n] || node)
        idx ← idx >> 1
k. roots[i] ← node
10. sk_last ← sig[offset : offset + n]
11. root_last ← sig[offset + n : offset + 2 × n]
12. roots[k-1] ← root_last
13. setTypeAndClear(ADRS, SL_FORS)
14. setTreeHeight(ADRS, a)
15. setTreeIndex(ADRS, 0)
16. pk ← Tw(PK.seed, ADRS, roots[0] || ... || roots[k-1])
17. return pk
```

5.5 SLH-DSA. The Stateless Fallback

5.5.1 Parameters

Parameter	SHRINCS-SHA2-128s
`h`	63 (total tree height)
`d`	7 (number of hypertree layers)
`h'`	9 (height per layer)
`a`	12 (FORS tree height)
`k`	14 (number of FORS trees)
`w`	256
`n`	128

5.6 Size

5.6.1 Stateful signature size

The WOTS+C component is 292 bytes: randomness (32 bytes) + counter (4 bytes) + chain values (256 bytes). The authentication path for signature `q` (count after signing) is `q * (n/8) = q * 16` bytes. The stateless public key `PK_sl` adds `(n/8) = 16` bytes.

$$\texttt{sig_sl} = 292 + q * 16 + 16 = 308 + q * 16 \text{ bytes},$$

where q is the signature count after signing:

q	Auth path	Total
1	16 bytes	324 bytes
2	32 bytes	340 bytes
3	48 bytes	356 bytes
10	160 bytes	468 bytes

5.6.2 Stateless signature size

5.7 SHRINCS Core Algorithms

5.7.1 Key Generation

```
Algorithm: SHRINCS.KeyGen()
Output:
    SK: secret key
    PK: public key
    state: initial state

1. seed ←$ {0,1}^384
2. SK.seed ← seed[0:16]
3. SK.prf ← seed[16:32]
4. PK.seed ← seed[32:48]
5. ADRS ← new_ADRS()

// Generate stateful public key (UXMSS root)
6. setLayerAddress(ADRS, 0)
7. setTreeAddress(ADRS, 0)
8. PK_sf ← compute_initial_stateful_root(SK.seed, PK.seed, ADRS)

// Generate stateless public key (SPHINCS+C root)
9. PK_sl ← sphincs_root(SK.seed, PK.seed, ADRS)

// Combined params
10. PK.root ← H(0x08 || PK.seed || PK_sf || PK_sl)
11. SK ← (SK.seed, SK.prf, PK.seed, PK.root, PK_sf, PK_sl)
12. PK ← (PK.seed, PK.root)
13. state ← { q: 0, valid: true }
14. return (SK, PK, state)
```

5.7.2 Recovery from Seed

```
Algorithm: SHRINCS.Restore(seed)
Input:
    seed: master seed
Output:
    SK: secret key
    PK: public key
    state: state "LOST"

1. SK.seed ← seed[0:16]
2. SK.prf ← seed[16:32]
3. PK.seed ← seed[32:48]
4. ADRS ← new_ADRS()
5. PK_sf ← compute_max_stateful_root(SK.seed, PK.seed, ADRS)
6. PK_sl ← sphincs_root(SK.seed, PK.seed, ADRS)
```

```
7. PK.root ← H_n(0x08 || PK.seed || PK_sf || PK_sl)
8. SK ← (SK.seed, SK.prf, PK.seed, PK.root, PK_sf, PK_sl)
9. PK ← (PK.seed, PK.root)
10. state ← { q: $\bot$, valid: false }
11. return (SK, PK, state)
```

5.7.3 Statefull signing

5.7.4 Stateless signing

5.7.5 Statefull verification

5.7.6 Stateless verification

5.8 Data structures

5.8.1 Secret Key Format

5.8.2 Public Key Format

5.8.3 Stateful Signature Format

5.8.4 Stateless Signature Format